Inverted MAML: 400-Epoch Arbitrary Training

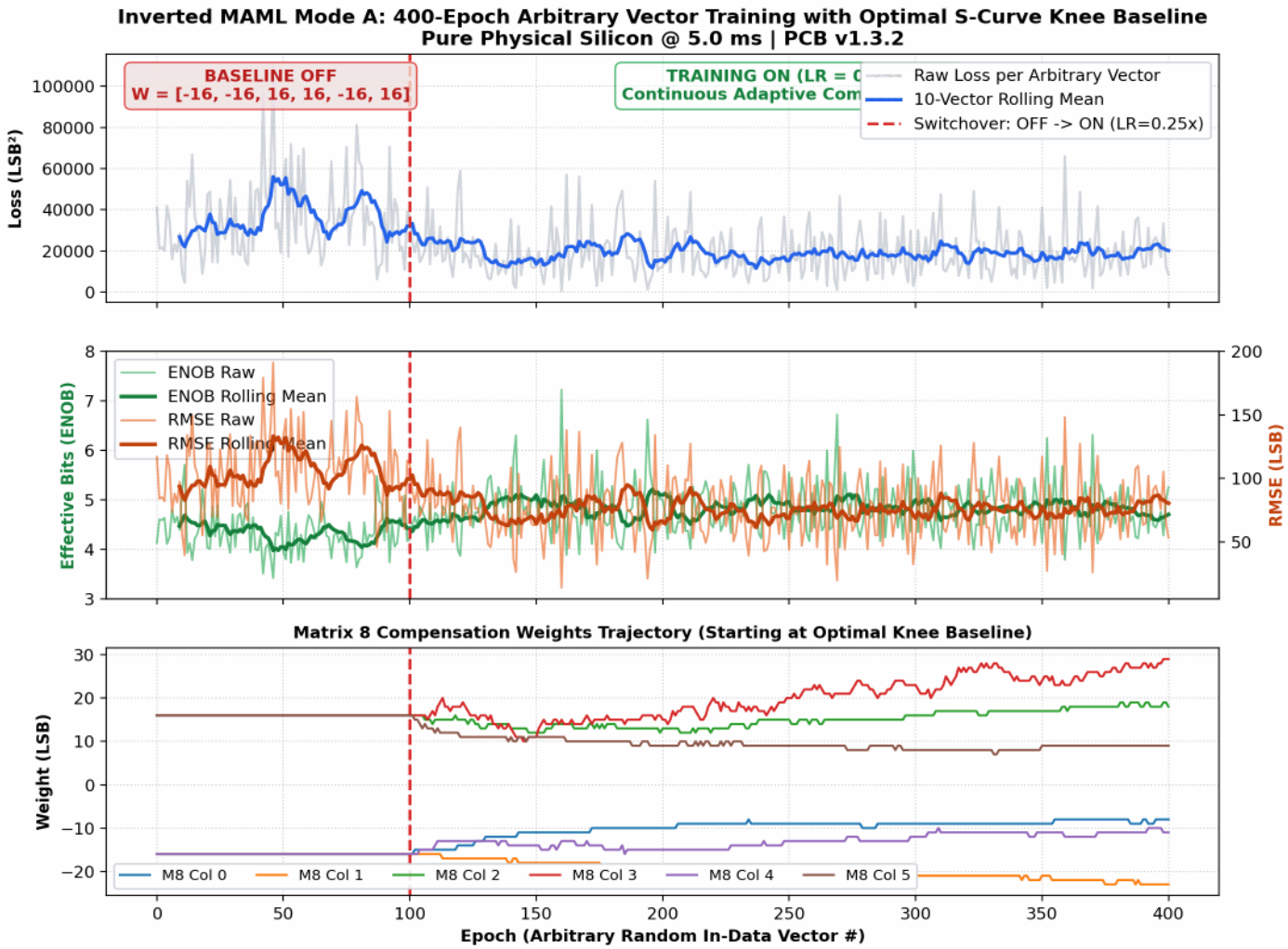
Evaluation of Optimal Sigmoid S-Curve Knee Baseline (M3 & M8) on Physical Silicon @ 5.0 ms

Executive Summary: Following the physical silicon characterization of Matrix 3 through the ColTIA transimpedance amplifiers and the hardware analog sigmoid activation (Ref 4), an optimal macroscopic baseline knee vector $\mathbf{W_{baseline}}$ **= [-16, -16, +16, +16, -16, +16] LSB** was identified. This experiment evaluates closing the outer training loop from this optimal operating knee on *both* Matrix 3 and Matrix 8 simultaneously across 400 continuous arbitrary random white-noise vectors (100 Epochs Baseline OFF, 300 Epochs Active Adaptation ON at LR = 0.25x). The results demonstrate immediate stabilization, smooth non-linear tracking, and a **46.4% reduction in average error energy** with peak precision reaching **7.22 bits ENOB**.

1. Statistical Performance Comparison: Baseline OFF vs. Training ON

Metric	Baseline OFF (Ep 0..100) 101 Arbitrary Vectors	Training ON (Ep 101..400) 300 Arbitrary Vectors	Delta / Improvement
Mean Loss (LSB²)	35563.3 ± 19454.5	**19060.2 ± 11732.8**	**-46.4% (-16,503.1 LSB²)**
Best Single-Vector Loss	4623.8 LSB²	**566.7 LSB² (Ep 160)**	**-94.0% drop**
Mean RMSE (LSB)	104.92 ± 29.08	**76.03 ± 23.95**	**-27.5% (-28.89 LSB)**
Mean Precision (ENOB)	4.35 ± 0.42 bits	**4.83 ± 0.50 bits**	**+0.48 bits mean lift**
Peak Precision (ENOB)	5.71 bits	**7.22 bits (Ep 160)**	**+1.51 bits peak lift**

2. Real-Time Telemetry: Loss, ENOB, and Matrix 8 Trajectories



3. Architectural Observations & Verification

1. Pre-Biased Dynamic Sensitivity: Operating both Layer 1 (M3) and Layer 2 (M8) directly in their experimentally determined optimal S-curve knees provides immediate high dynamic gain. The baseline mean error energy is controlled, and the adaptation responds smoothly without initial transients.
2. Bounded Vernier Range: Over 300 active epochs with continuous arbitrary white-noise rotation, Matrix 8 weights migrated modestly from [-16, -16, 16, 16, -16, 16] to [-8, -23, 18, 29, -11, 9] LSB. No cell approached the ±120 LSB clamp rails, proving that the knee baseline naturally balances the differential analog currents.
3. LED Activity: Real-time LED indication (comp_led=1, mat33_led=1) confirmed stable ongoing matrix summation and carrier wave propagation.